IE 6400 Project 1

Los Angeles Crime Analysis

Authors: Anika Mamidwar, Sanya Kaluarachchi, Jogeashwini

Abstract

In this project, we are examining crime data from Los Angeles from 2020 to 2024. Through different analyses, we identified trends, seasonal patterns, and demographic relationships among reported crimes. After taking a public data set from Data.gov, we cleaned any inconsistencies, standardized variables, and removed unnecessary columns. Using visualization methods, we conducted exploratory data analyses to find trends in yearly, monthly, and weekly crime patterns as well as demographic and regional influences on crime rates. Our results highlight shifts in crime counts based on political and public health events. We identified noticeable declines in crimes during the time of the pandemic. These results provide a structured overview of crime trends in Los Angeles and highlight the importance of data-driven approaches in addressing urban crime.

Data Inspection

The following list of variables is used by the LAPD to maintain the crimes reported dataset. Further descriptions of the analysis performed and manipulation done to variable categories are described below in the following sections.

Variable Name	Variable Meaning
DR_NO	Division of Records Number: Official file number made up of a 2-digit year, area ID, and 5 digits
Date Rptd	Recorded date the crime was reported
DATE OCC	Reported date that the crime occurred
TIME OCC	Time crime reported
AREA	The LAPD has 21 Community Police Stations referred to as Geographic Areas within the department. These Geographic Areas are sequentially numbered from 1-21.
AREA NAME	The 21 Geographic Areas or Patrol Divisions are also given a name designation that references a landmark or the surrounding community that it is responsible for
Rpt Dist No	A four-digit code that represents a sub-area within a Geographic Area
Part 1-2	
Crm Cd	Indicates the crime committed. (Same as Crime Code 1)
Crm Cd Desc	Defines the Crime Code provided.
Mocodes	Modus Operandi: Activities associated with the suspect in the commission of the crime.
Vict Age	Age of the individual reporting the crime
Vict Sex	Sex of the individual reporting the crime
Vict Descent	Ethnic descent of the individual reporting the crime
Premis Cd	The type of structure, vehicle, or location where the crime took place.
Premis Desc	Defines the Premise Code provided.
Weapons Used Cd	The type of weapon used in the crime.
Weapons Desc	Defines the Weapon Used Code provided.
Status	Status of the case.
Status Desc	Defines the Status Code provided.
Crm Cd 1, Crm Cd 2, Crm Cd 3, Crm Cd 4	Indicates the crime committed. Crime Code 1 is the primary

	and the most serious one. Crime Code 2, 3, and 4 are respectively less serious offenses. Lower crime class numbers are more serious.
Location	Street address of crime incident rounded to the nearest hundred block to maintain anonymity
Cross Street	Cross Street of rounded Address
LAT	Latitude
LON	Longtitude

Data Cleaning

When acquiring a large data set, the first and most important step is to clean the data set to help correct any inconsistencies and remove unnecessary data. After reviewing each variable and its row entries, we realized there were many fixes we needed to make. First, we noticed the gender column had M/F entries for male and female, and the other entries are under X, '-', and H. We standardized the column entries so each non M/F entry would be an X, which helped maintain consistency.

We have two columns, "Date Reported" and "Date Occurred," that track the dates of the crimes. We took these two columns and converted them to a date-time format. By converting it, we can easily extract the specific day, month, or year from each of the dates, which is useful for future analyses.

We removed the columns: Weapon Description, Weapon Used Code, Mocodes, and Cross Street. Although this could be very useful when doing other analyses, it was not beneficial to us.

As we continued to create our visualizations, we came across some unusual observations, which we confirmed were due to the 2025 data being included. Because data from 2025 was only recording up to March, it was messing up calculations when counting total crimes year to year. Removing the data helped make sure we weren't seeing unusual outcomes in our visualizations.

Data Analysis

Figure (1) portrays the raw count of documented crimes in Los Angeles from the period of 2020 to 2024. Crime steadily increases subsequent to 2020, peaking in 2022. This could be

coinciding with the dropping of COVID-19 pandemic related restrictions as well as economic turmoil with businesses still recovering subsequent to the pandemic, as well civil unrest across the country from a change in presidential administration early into 2021, midterm elections and congressional changes in 2021 to 2022, supply-chain upheaval with the beginning of the war in Ukraine, wildfires across California [1], amongst several other political and social upheavals [2].

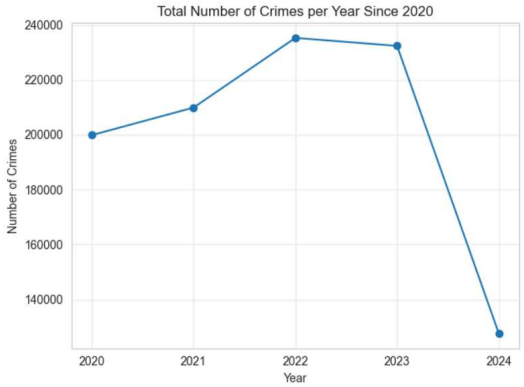

Figure 1. Total number of crimes within LA reported to the LAPD from 2020 to 2024.

It is notable that 2024 shows a marked reduction in crime recorded, which whilst has several policy choices might have influenced [3], the most likely explanation is that the LAPD adopted a new record management system in compliance with FBI regulations March of 2024, with fewer incidents reported in the existing system which this dataset is being pulled from. This explains the sudden decrease associated with monthly total crimes as depicted in Figure (3) that occurs in 2024 of March and continues to steadily decrease throughout the year as the old system is phased out.

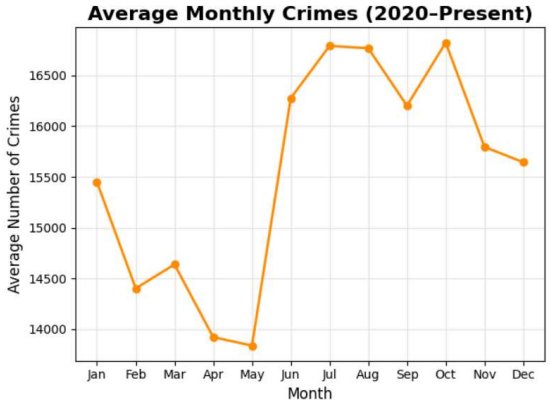

Figure 2. The number of crimes occurring per month averaged across a 4 year period.

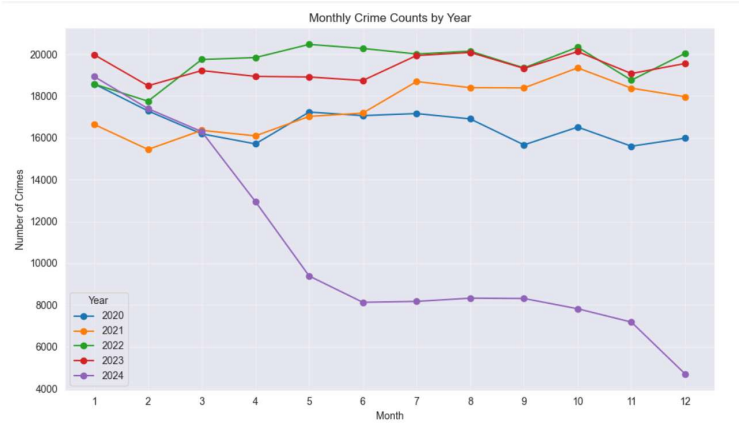

Figure 3. The total monthly crime counts over the course of 2020 -2024 period

In addition to temporal and raw numerical counts, analysis was done to better understand the nature of crimes committed. Figure (5) illustrates the prevalence of vehicular theft, with homicides making up just a small percentage of the crimes reported in the four year period.

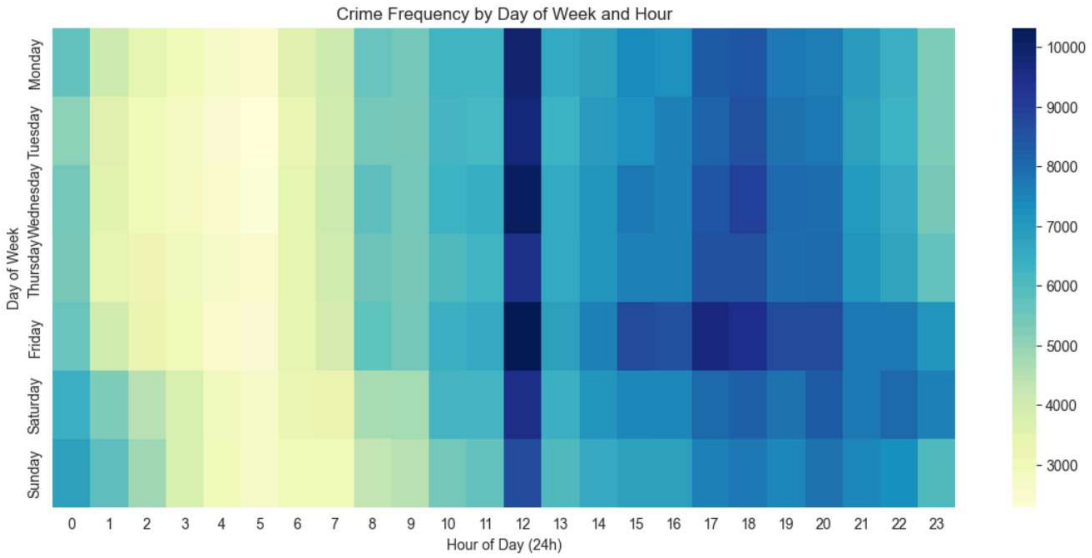

Figure 4. The influence of temporal status - time of day and day of the week on crime prevalence as depicted through a heatmap, with higher crime counts depicted in darker more saturated blue shades and lighter, less prevalence times taking on a lighter yellow-ish hue.

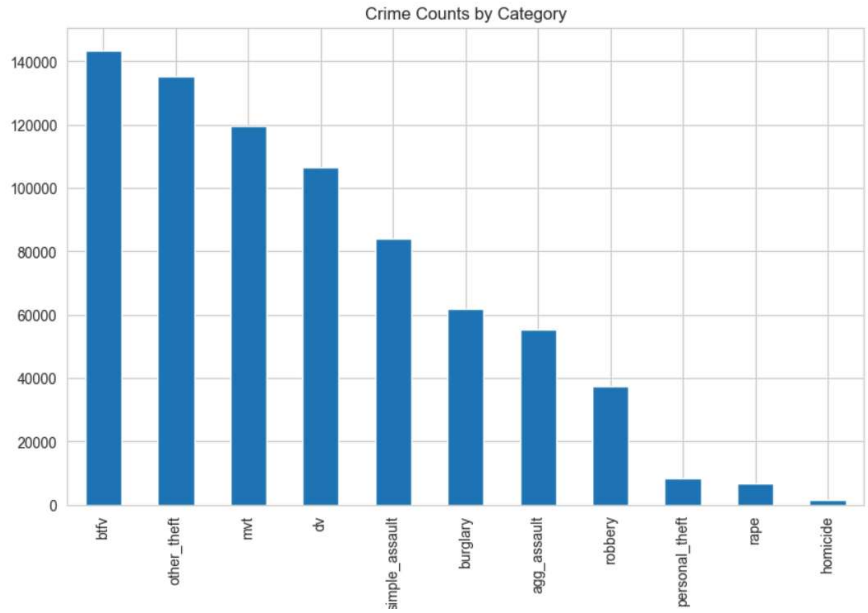

Figure 5. Crimes committed as distinguished by various crime categories including Burglary/Theft From Vehicle (btfv), all other theft distinctions which include shoplifting and thefts under certain monetary amounts (other_theft), motor vehicle theft (vt), domestic violenct (dv), simple assault (simple_assualt), burglary, aggravated assault (agg_assualt), robbery, personal theft which includes things such as pickpocketing or purse theft (personal_theft), rape, and homicide.

Additionally, the LAPD further subdivides crime reporting into the categories of violent crimes, which consists of charges related to homicide, rape, robbery, aggravated assault, domestic violence, and simple assault, and property crimes, consisting of burglary, motor vehicle theft, Burglary/Theft From Vehicle (btfv), personal theft, and all other theft. As depicted in Figures (6) and (7), there are similarities in the trend of higher crime associated with the weekend, versus lower crime building up to this point through the weekdays. However, whilst in almost all years violent crimes peaked on Sunday, property crimes peaked on Friday throughout the years.

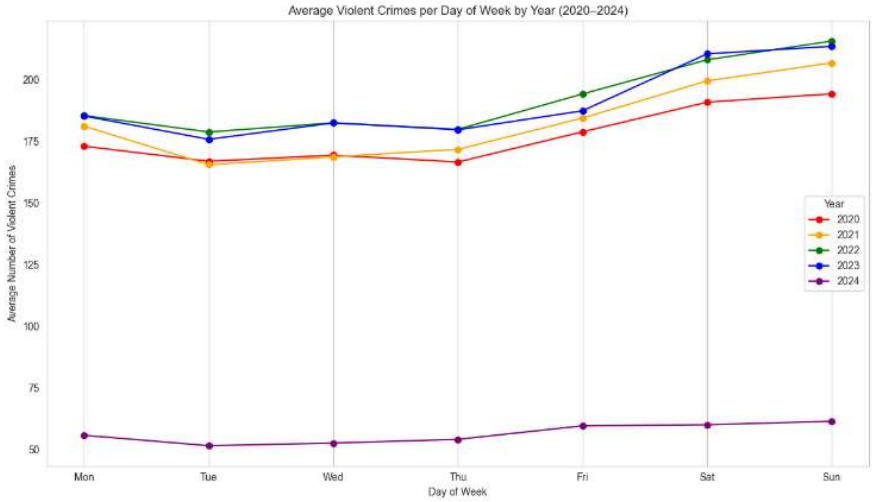

Figure 6. Depicts the average occurrence of violent crimes per week day. Violent crimes is as defined as all homicide, rape, robbery, aggravated assault, domestic violence, and simple assault charges.

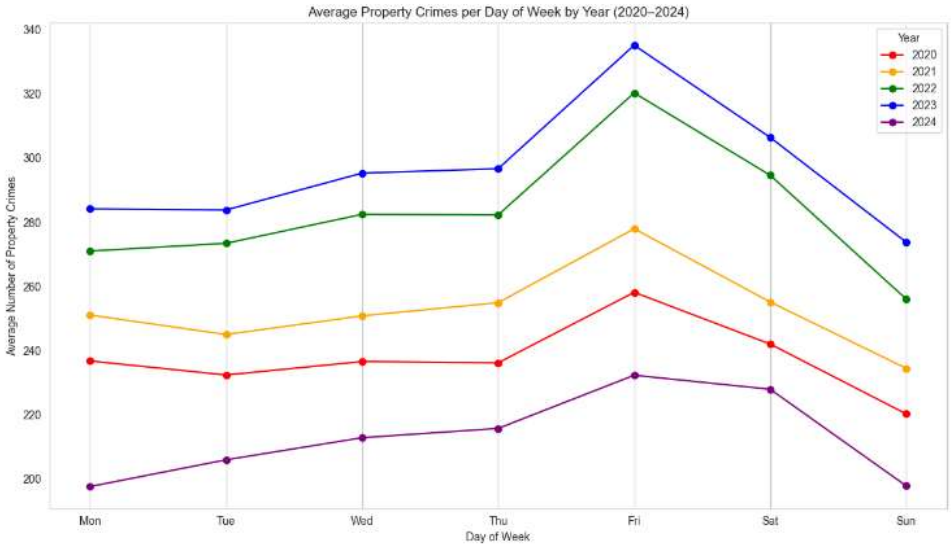

Figure 7. Depicts the average occurrence of property crimes per week day each year. Property crimes is as defined as all burglary, motor vehicle theft, Burglary/Theft From Vehicle (btfv), personal theft, and all other theft charges.

LA County is composed of approximately 100 different cities, and even more divisions when accounting for unofficial area distinctions [4]. As depicted by the heatmap in Figure (8), crime stays fairly consistent throughout the cities, with a notable exception of both 77th Street and Central, which remains the highest crime count categories. Figure (9) depicts the raw count of each area as well.

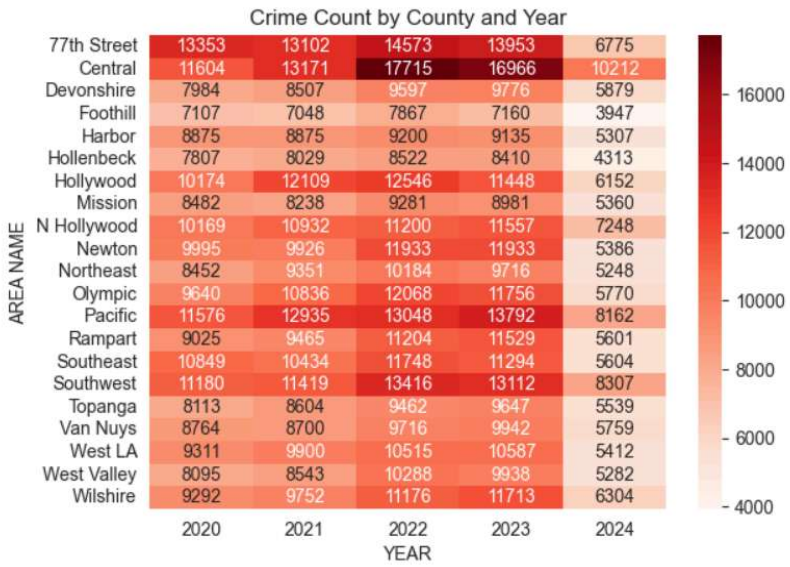

Figure 8. A heatmap indicative of reported crime incidences throughout the four year period as distinguished by different report geographical locations. Darker hues indicate a higher crime count.

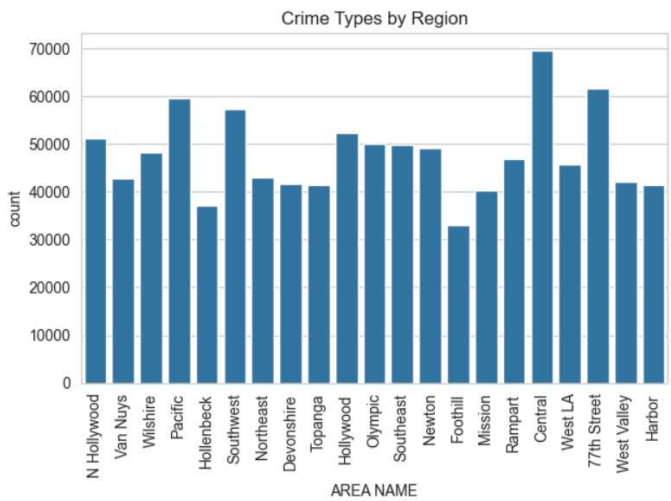

Figure 9. A histogram indicative of the reported crime incidences reported throughout the four year period as distinguished by different report geographical locations.

Finally, analysis was done to better understand the relationship between Victim Sex and the relationship to crime. Figure (11) depicts how reported crime is broken down by the filing party's sex, and what percentage of crimes each sex characteristic is said to make up for the annual reported crime. 2024 indicates the largest population of crimes associated with the sex characteristic 'Unknown' - which could once again be traced back to the

aforementioned data anomaly with the old reporting system, which the dataset is drawn from, failing to maintain the most up to date information.

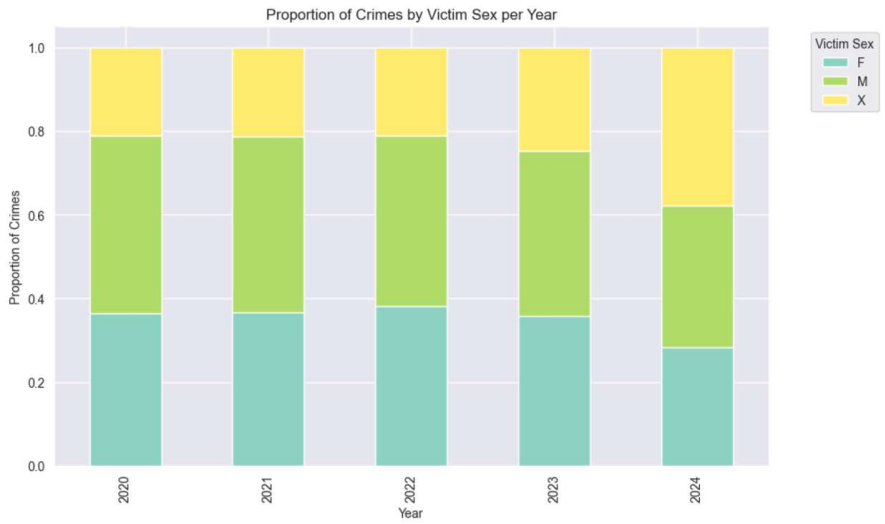

Figure 10. The attached histogram identifies the sex differentiation in reported crimes each recorded year, with blue identifying crimes with victims classified as 'Female', green classifying 'Male', and unknown Victim sex identified with 'Yellow'.

Discussion

The prior data analyses were done to portray the crime characterizations of the City of Los Angeles between 2020 and 2024, illuminating potential socioeconomic and political factors that may influence crime in the area. For example, as seen with the decline in 2020 declining as the city was forced into lockdown due to the COVID-19 pandemic and thus mobility was restricted with social distancing and business closures, likely reducing opportunities for both property and violent crimes. The return to policies that allowed for increased socialization and mobility in 2021 and 2022 was also marked by great social tensions throughout a changing political and economic landscape, evidently characterized by mass civil unrest and reported crime.

On a more temporal scale, there seems to be clear patterns regarding the cyclical nature of crime in a week and even throughout the year, with property crimes consistently peaking on Fridays, violent crimes peaking on Sundays. This could certainly be related to increased social presence as the typical 9-5 work week gives way to the weekend. Additionally, our time-of-day analysis portrays how crime typically occurs later in the day, from the later

afternoon to evening hours, which is indicative of once again likely increased population and social interactions as more individuals are outside for outdoor and commercial activities. These insights into time-related understandings can help to influence the distribution of law enforcement throughout the day, and also influence initiatives aiming at preventive efforts.

Given the aforementioned anomaly with the dataset's lack of reporting for the year of 2024 and the lack of entries past March of 2025 in the dataset, we were interested in exploring the use of predictive analytics to compensate for the missing data entries utilizing prior monthly trends from 2020 - 2023. Utilizing time-series modeling with the Python package Prophet, forecasting was performed. As seen in Figure (10) below, there is an overall trend for the dataset to decrease subsequent to 2020, which continues on in the prediction to both 2024 and 2025. However, these forecasting efforts do not account for a changed political landscape nor any legal reforms or socioeconomic factors aside from those accounted for by prior years. Thus, future predictions could be improved by incorporating socioeconomic factors and the suspected influence of legal and political reforms.

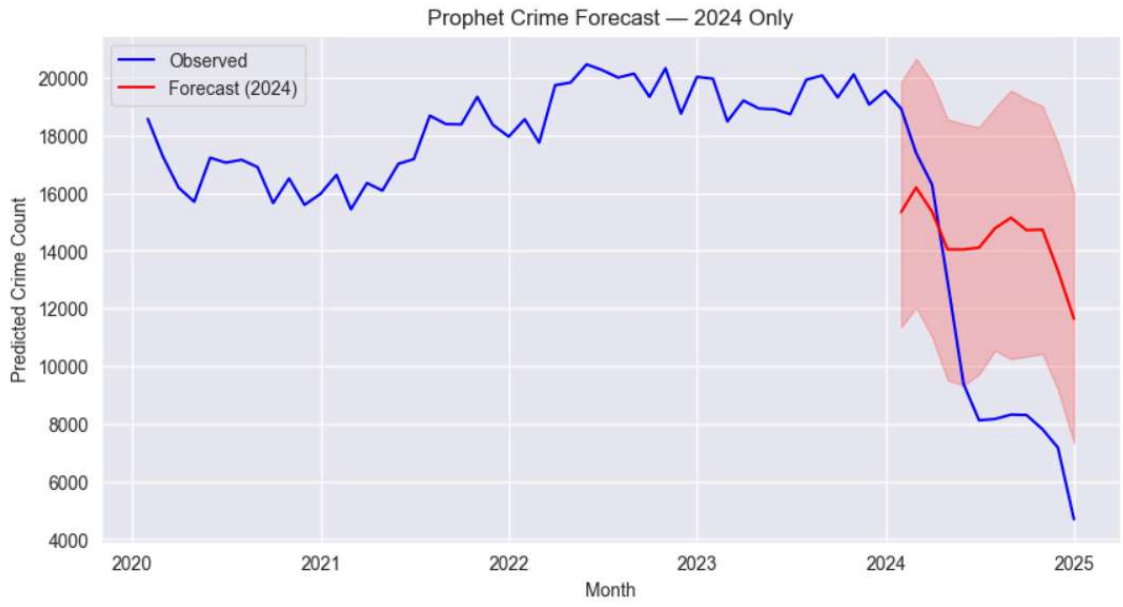

Figure 11. The figure depicts the predicted reported crime for 2024-2025, portraying the forecasted region with a Red line of best fit, adding a surrounding highlighted region for the margin of error. The blue line depicts the actual record of crime data from the dataset.

Some limitations of our analysis include the influence of underreporting. While we acknowledged 2024-2025's missing data, there are particularly anomalies expected for crimes related to domestic violence and sexual assault, which despite their surveyed prevalence [5], have very low reporting rates likely due to social stigmas. To help better mitigate these data inconsistencies, it would be necessary to cross reference these datasets with other available public records.

Overall however, the findings highlight the complexities ruling Los Angeles crime patterns, from social behaviors, geographical location, and potentially other political and systemic criminal reporting practices. In the future, using socioeconomic and environmental factors to better understand how they influence crime's prevalence will allow data-driven insights to guide more equitable and effective legal reforms, community initiatives, and resource allocation within the city of Los Angeles.

Conclusion

Works Cited

[1] MacCarthy, J., & Richter, J. (2025, February 5). 4 graphics explain Los Angeles' rare and devastating January fires. World Resources Institute. Retrieved from https://www.wri.org/insights/los-angeles-fires-january-2025-explained

[2] These were the top political stories of 2022. (2022, December 31). NPR. Retrieved from https://www.npr.org/2022/12/31/1146261338/2022-political-stories-midterms-abortion-inflation-immigration-ukraine?

[3] Mayor's Office, City of Los Angeles. (2023, December 7). LAPD reports homicides and violent crime down in 2023 as Mayor Bass highlights comprehensive strategy to keep Angelenos safe. Retrieved from https://mayor.lacity.gov/news/lapd-reports-homicides-and-violent-crime-down-2023-mayor-bass-highlights-comprehensive

[4] County of Los Angeles. (n.d.). Maps & geography. Retrieved [date you accessed] from https://lacounty.gov/government/about-la-county/maps-and-geography/

[5] The National Domestic Violence Hotline. (n.d.). Domestic violence statistics. Retrieved [Access Date] from https://www.thehotline.org/stakeholders/domestic-violence-statistics/